

Virtual Reality and Creativity: How do Immersive Environments Stimulate the Brain during Creative Idea Generation?

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Figure 1: Participants performed a creative idea generation task while wearing a VR headset and being immersed in a virtual environment.

ABSTRACT

Virtual Reality (VR) environments are promising creativity support tools that can be designed to enhance factors related to the creative process such as immersion, engagement, and flow. However, it is still unclear how brain responses associated with creativity are influenced by VR. To address this gap, we collected EEG signals from 21 participants during a creative idea generation task in two conditions; 2D Screen vs. 3D VR. The EEG power in the alpha (8-13 Hz), beta (13-30 Hz), and gamma (30-45 Hz) frequency bands were compared. The results showed that participants experienced higher enjoyment, immersion, and temporal dissociation in the 3D VR condition. Additionally, a significant increase of oscillations in the high-frequency bands of beta and gamma was observed in VR. The findings suggest that VR can induce a more pronounced sense of flow that is associated with distinct brain activation patterns.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI.**

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KEYWORDS

Creativity, Brain activity, EEG, Virtual Reality, Immersion, Flow

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1 INTRODUCTION

Creative idea generation, the ability to produce ideas that are original and useful [28], is a crucial twenty-first century skill. Due to its fundamental role in innovation, it directly impacts scientific, economic, and social domains [15]. Such creativity is necessary when no apparent satisfactory solution to a problem is in view. Here, the generation of original ideas is needed to break free from an existing sub-optimal situation, in favor of a potentially better one. Emerging technologies can enable new ways in which creative idea generation is facilitated and enhanced, and the development of such digital creativity support tools has been hailed as a 'grand challenge' in the field of human-computer interaction [25]. One technology that offers a particularly promising platform for supporting creative idea generation is Virtual Reality (VR) [15, 18, 27, 32].

VR is an artificial 3D simulation of an environment and is believed to engage cognitive processes similar to how people experience the real world [16]. The potential of VR systems has been

investigated in various fields such as education [10, 23], training [22, 36], and entertainment [10, 24]. However, relatively little is known about how VR can be used to support creative idea generation [15, 35]. One study by [18] suggests that the flexibility of VR for creating environments that are not constrained by real-world physical limitations enables new forms of creative expression. Others have suggested that VR can also support creative idea generation because of its effect on the flow experience [20], which subsumes concepts such as immersion [21], enjoyment, and distorted time perception [29]. Specifically, [35] showed how designing wearable products in VR, compared to using pen-and-paper, induced more flow and resulted in more creative ideas. Similarly, [32] showed that receiving design instructions in VR, rather than during a lecture, could better support creative idea generation due to its positive effect on immersion.

A currently understudied use of VR for supporting creativity is its potential for allowing the user to "be in" and "explore" an environment that otherwise would not be accessible [32]. For example, when generating creative ideas to save Venice from tourism, it can be useful to explore Venice itself, which might not be a viable option in real-life. VR can thus facilitate the production of creative ideas for solving problems that are bound to the physical environment. This often requires sensory exploration and integration of information from the surroundings to generate and evaluate new ideas [27, 32]. The immersive feature of VR enables the user to search for new information in the environment [26] while experiencing an effective sensory-motor coupling enhanced by the experience of flow, especially when compared to more traditional digital tools such as 2D screens [35].

To effectively support creative idea generation through VR, it is crucial to monitor users' performance and mental states, such that VR-based interventions can be designed to provide nudges and help people to maintain a state that optimally supports their creative thinking. The first step in the development of such closed-loop VR systems is to investigate the potential of an objective measurement that can be tracked in real-time. Currently, assessments of creative idea generation are often conducted through self-reports or ratings by domain experts, and both measurements are highly subjective and only possible after the completion of the task [4]. An alternative approach is to record brain activity during creative tasks to examine underlying neurophysiological indicators of creative idea generation [6, 13, 30]. Electroencephalography (EEG) is a particularly suitable measure for this purpose due to its high temporal resolution [30] as well as its accessibility and potential to be combined with VR.

While several studies have previously investigated the neural mechanisms of creative idea generation [6, 13, 30], very few have done this in combination with immersive VR environments. These studies often reported a positive correlation between alpha band activity (8-13 Hz) and creative task performance [13, 30]. However, more recent studies employing EEG during creative tasks that demand externally oriented attention rather than internal focus, report brain activity changes in higher frequency ranges such as beta (13-30 Hz) and gamma (30-45 Hz) frequency bands [30, 32]. Therefore, this study aimed to identify the brain activity components that are affected by creativity support tools such as immersive VR, which facilitate visual exploration of a simulated environment.

We recorded the EEG brain activity of 21 individuals who worked on a creative idea generation task while interacting with a 360° virtual environment under two levels of immersion: when the environment was presented on a 2D Screen vs. when the subject viewed it in 3D using a VR head-mounted display. Using this approach, we aimed to answer the following research question:

RQ: Is there a difference in alpha, beta, and gamma brain activity when people engage in creative idea generation in a virtual environment presented in 3D VR compared to a 2D screen?

2 METHODS

2.1 Participants

Twenty-one university students (15 females, 6 males, $M_{age} = 20.35$, $SD_{age} = 2.34$) participated in this study. All participants reported no or very little experience with VR. They received information about the study and consequently signed an informed consent form in accordance with the guidelines of the Research Ethics and Data Management Committee of Tilburg University.

2.2 EEG recording

Brain activity was measured using the g.Nautilus EEG equipment (g.tec Medical Engineering, Austria). The EEG signals were recorded via 16 electrodes according to the international 10–20 system (Figure 2). In this paper, we only report the measurements from the frontal brain area (i.e., F3 and F4 electrodes, shaded red in Figure 2). The recording sampling rate was 250 samples per second. To minimize noise, a 48-52 Hz Notch filter was applied.

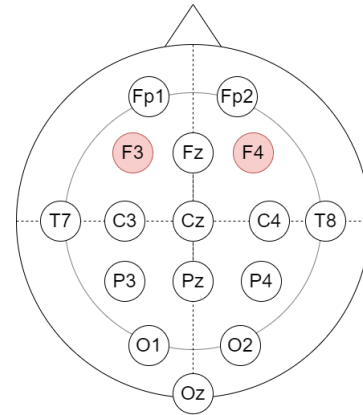


Figure 2: EEG brain activity was recorded by 16 electrodes distributed across the scalp. In this paper, the signals from the frontal electrodes F3 and F4 were analyzed.

2.3 Conditions

In this study, the experiments were conducted using a within-subjects design, where participants experienced two experimental conditions in a counterbalanced order.

2D Screen Condition. In the Screen condition, participants sat in front of a 27-inch monitor with a 1920 × 1080 resolution (Full HD) and interacted with a 360° video displayed on the screen.

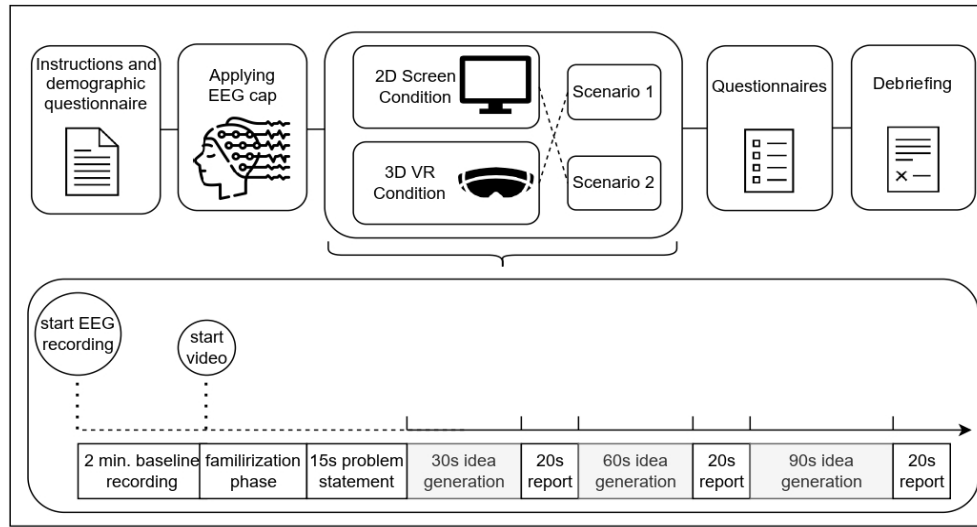


Figure 3: The experiment protocol.

The participants were able to change perspective and navigate the scene using a computer mouse but were not able to move or change their location in the video. Before the task started, EEG signals were collected in a baseline phase, in which participants watched a fixation cross on the screen.

3D VR condition. In the VR condition, participants wore a VR head-mounted display (Oculus Quest) and watched a 360° video (a different video than the other condition, also counterbalanced) in the 3D environment. Again, the participants were not able to change their location within the 3D environment but could move their head to look around and see the environment from different perspectives. Before the task, the same fixation cross as in the 2D screen condition was shown in the VR environment to record the baseline EEG while subjects were wearing the VR headset. This was to offset any potential noise that might have been registered in the recordings due to the placement of the headset. In the past, the effect of immersive VR was compared to a non-immersive pen-and-paper control condition [32]. However, in this study, we were interested in the effect of technology type as a creativity support tool. The 360° videos used in the 2D Screen condition were the same as the ones used for the 3D VR condition. Therefore, by comparing the 2D and 3D conditions, we intended to increase the reliability of measuring the effect of VR immersion on creative idea generation.

2.4 Creative idea generation task

A common approach to test creative idea generation is to present participants with a problem statement and to ask them to generate as many creative ideas that might help solve the problem as they can [6, 30]. The task used in this study started with a problem statement that was presented verbally and was related to the observable scenery in each condition. Subjects received time to generate creative ideas in response to the problem statement while exploring the environment.

Two problem statements were prepared for this research both related to climate change issues. According to [7], creativity requires

a certain level of knowledge on a specific topic. Because of this, we chose a topic that we assumed participants were more or less equally informed about.

Scenario 1: In the first scenario, participants were shown a first-person view of a forest. The problem statement described that the forest is situated in the middle of two villages that want to engage in trade. The participant was asked to come up with ideas that enabled transport between the villages without destroying the forest.

Scenario 2: In the second scenario, participants were shown a first-person view of an island. It was stated that a rare species of lizards are living on this island, but that they are endangered by the rising sea levels. The participant was asked to come up with ideas for saving the lizards from going extinct.

While choosing the two 360° videos¹ for this experiment, we mainly focused on visual content that included sufficient stimuli to enhance the creative thinking processes without overstimulating the participant through excessive movement or distracting information that is irrelevant to the problem statement (e.g., a forest with several trees and bushes in minimal movement, but no humans or animals.)

2.5 Procedure

Before the study, each participant received an information letter that held information about the general setup of the experiment. After signing the informed consent form, they provided their demographic information. Next, they were seated in a comfortable chair in front of a computer screen to be prepared for EEG recording. The experimenter positioned the EEG cap on the participant's head and applied conductive gel to improve the quality of signals.

¹Both videos were downloaded from <https://www.storyblocks.com/>. Permission to download and use the videos in the experiment was granted through a subscription to an individual license. Links to previews of the videos can be found in the appendix. Please note that these are watermarked versions as we were not permitted to distribute the original videos.

The experiment started with one of the two conditions (2D Screen or 3D VR) in a counterbalanced order. Each condition entailed 3 phases: 1) a familiarization phase in which participants explored the environment before the task to inhibit the novelty effect, 2) an EEG baseline recording (2-min) gathering each individual's brain signals in the resting state before the task, and 3) the creative idea generation task, in which participants were asked to come up with creative ideas that help solve the stated problem.

To reduce EEG artifacts generated by jaw movement, participants were instructed to hold ideas in their memory until they were signaled to report them. Each condition was segmented into 3 idea generation phases and 3 report phases (Figure 3). During the report phases, participants could "unload" their memory without disturbing the EEG recording. The reporting of ideas took place in 3 intervals of 20 seconds. The idea generation phases took place in three uneven parts lasting 30, 60, and 90 seconds, respectively. Therefore, the task in each condition lasted 4 minutes, consisting of a total of 3 minutes of creative idea generation and a total of 1 minute of idea reporting.

These intervals were chosen based on several considerations. For instance, it has been shown that as participants iterate ideas in divergent thinking tasks, the originality of their ideas increases over time [3]. Besides that, it has been shown that time pressure can negatively affect creative idea generation [3]. Finally, the exact amount of seconds most appropriate for the segments was determined through pilot testing².

After each condition, the participants were instructed to fill in a questionnaire about their experience.

2.6 Evaluation and Analysis

To evaluate the impact of immersive environments on creative idea generation, we compared the two conditions (3D VR vs. 2D Screen) using subjective and objective measures. For subjective measures, we obtained participants' scores on Temporal dissociation, Immersion, and Enjoyment following [1]. Immersion refers to the experience of total engagement in the task and was measured using 4 items. Enjoyment refers to the pleasurable aspects of the interaction and was measured using 5 items. Temporal dissociation, which entailed 5 items, refers to distorted time perception and the inability to register the passage of time, i.e., how fast time flew during the interaction. The item scores were aggregated per construct and compared between conditions. After the data distribution was checked for normality using the Shapiro-Wilk test, a paired t-test was applied to examine whether the scores on Temporal dissociation, Immersion, and Enjoyment were significantly different between the two conditions. In addition to this, Cohen's *d* was used to measure the effect sizes. The analysis was performed in R.

As for the objective measurement, we compared the neural responses of participants between the two conditions. Following the literature on the neuroscience of creativity [30, 32], we were particularly interested in EEG signal power changes in the frontal area of the brain in the frequency bands of alpha (8 - 13 Hz), beta (13 - 30 Hz), and gamma (30 - 45 Hz) when participants engaged in the idea generation task using either 3D VR or 2D screen. The recorded EEG signals were first pre-processed using a bandpass

filter and Independent Component Analysis (ICA) to clean the data from artifacts and eye blinking. Next, the two frontal electrodes F3 and F4 were selected and the time series from these two channels were segmented in each condition to acquire the "pre-task baseline" and the "creative task" epochs. The signal power in the target frequency bands was calculated separately for baseline and task epochs using the spectopo function from EEGLAB [12]. Then, for each participant in each condition, the EEG band powers in the baseline were subtracted from the corresponding power in the task phase. This baseline-correction approach was implemented to address the inherent individual differences in the EEG signals and hence obtain the task-induced changes in the dependent variables for each subject separately. This approach eventually created 3 frequency band power values (alpha, beta, gamma), per electrode (F3 and F4), per condition (3D VR vs. 2D Screen), and per subject. The values were then compared between 2D Screen and 3D VR conditions for each frequency band at each EEG site (i.e., electrode) separately. Since the obtained data did not follow a normal distribution, a paired Wilcoxon signed-rank test was applied. Also, since the comparison of band powers was repeated two times for the selected EEG channels (repeated measures), the adjusted p-value for significance was set at $.05/2 = .025$.

3 RESULTS

3.1 Questionnaire

Figure 4 displays the distribution of questionnaire scores in the two conditions. The paired t-test showed that participants experienced a significantly higher Temporal dissociation ($t(20) = 4.9314$, $p < .001$, $d = 1.1585$), Immersion ($t(20) = 5.6934$, $p < .001$, $d = 1.3662$), and Enjoyment ($t(30) = 5.5247$, $p < .001$, $d = 1.3103$) during the 3D VR condition than during the 2D Screen condition.

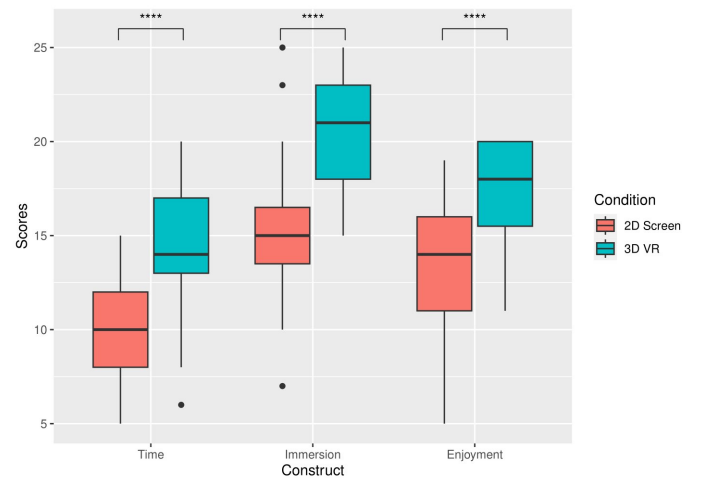


Figure 4: Questionnaire outcomes. Subjects reported significantly higher Temporal dissociation, Immersion, and Enjoyment when performing the idea generation task in 3D VR compared to the 2D Screen (** $p < .0001$).**

²Subjects that participated in these tests did not participate in the experiment.

3.2 EEG data

The distribution of the EEG frequency band power changes (from resting-state baseline) in the frontal brain region is demonstrated in Figure 5. The paired Wilcoxon signed-rank test showed a significant difference in beta and gamma band activity between conditions in both F3 and F4 electrodes. When subjects performed the idea generation task in the 3D VR condition, there was a significant increase of beta ($V = 31$, $p = .0022$) and gamma activity ($V = 20$, $p = .00035$) in F3 as well as in F4, where beta ($V = 29$, $p = .0016$) and gamma ($V = 8$, $p = .00002$) band power increased significantly compared to the 2D screen condition. The power changes in the alpha band were not significant in F3 ($p = .029$), nor in F4 ($p = .502$) at the adjusted significance level of .025 according to the Bonferroni correction [33].

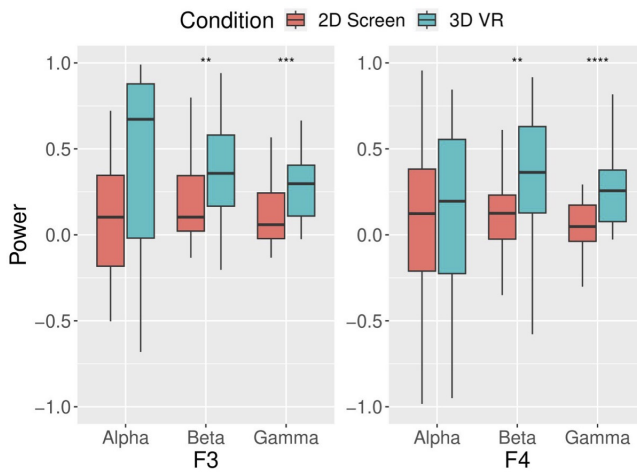


Figure 5: Distribution of EEG band powers. There was a significant increase of beta and gamma band activity (compared to baseline) in both F3 and F4 electrode locations when subjects completed the idea generation task in 3D VR compared to when they used 2D Screen ($p < .01$, *** $p < .001$, **** $p < .0001$).**

4 DISCUSSION

This research aimed to investigate the difference in brain activity when people performed a creative idea generation task in problem-related immersive virtual environments. EEG brain activity was recorded from 21 participants who generated creative ideas for a problem related to environmental preservation. During the task, participants could watch an immersive video of the environment related to the problem (e.g., a forest) using either a 3D VR headset or a 2D computer screen. The results showed that participants experienced significantly greater feelings of immersion, enjoyment, and time distortion, with a large effect size, when completing the task in the 3D VR condition than in the 2D screen condition. Additionally, the analysis of the EEG data showed significant increases in frontal beta and gamma band activity in the 3D VR condition. These findings suggest that VR can induce a more pronounced sense of

flow with distinct brain activation patterns that are associated with heightened attention and sensory processing.

These results are consistent with observations of [32] who also reported increased beta and gamma band activity in the subject group that received VR-based design instruction compared to the group that experienced lecture-based design training. Beta and gamma frequency bands represent *fast* brain responses that are generally associated with a state of alertness, outward attention, and high-level information processing [30]. Unlike alpha band activity that has been extensively reported as a common marker of creative thinking [13, 30], the correlation of beta and gamma frequency band increases with creative thinking is less commonly found. This is perhaps because most tasks designed to test creative idea generation in past studies draw strongly on internal cognition, e.g., memory search and mental imagery, to form relevant and coherent ideas. This contrasts with creative drawing or problem-solving tasks that are bound to the external environment, and rely on actively searching the environment to support idea generation. The high-speed neural processing in the gamma frequency band is considered a reflection of sensory processes and integration of information into conscious awareness [30], which are both relevant for such creative idea generation. Additionally, increased activity in the gamma band has been associated with the state of flow [2] and aha! moments [9]. This suggests that perceptually richer environments such as VR can facilitate idea generation through attentional and multi-sensory integration processes that help the user to be “in the zone”.

The findings of brain activity changes are complemented by the subjective reports, which revealed a significant difference in immersion, enjoyment, and temporal dissociation in the 3D VR condition. Feeling immersed is a central characteristic of a VR system, and the results show that the participants felt significantly more immersed in the 3D VR condition than in the 2D screen condition, which is in line with prior research [5, 17]. Previous work suggests that immersive VR can positively impact the users’ motivation, engagement [34], and enjoyment [21], which are factors known to have a positive effect on creative idea generation [8]. Time distortion, i.e., the feeling of time flying, is seen as a characteristic of the flow state [11, 29] and is closely linked to heightened enjoyment [29]. Indeed, all participants reported their experience in 3D VR as significantly more enjoyable than in the 2D Screen condition. But note that this could also relate to the novelty effect.

It has been established in previous HCI literature that the interaction environment is an essential factor for creativity support tools [26]. Creative activities rely on motivation, knowledge resources, social interaction, and exploration of new media technologies [14]. Virtual reality offers a rich 3D immersive experience where users can freely brainstorm, visualize ideas and develop prototypes in a cost-effective manner [15]. Moreover, the context and design of the virtual environment is another key factor that can be used to support creativity [26]. VR represents a “safe space”, where individuals can make mistakes and therefore explore their ideas without consequences [26]. It can also be designed to provide clues (i.e., additional pieces of information) and nudges to support creative idea generation [15]. By monitoring the brain activity of users in real time, it is possible to provide these clues and nudges in a more

timely fashion that would activate internal representations and stimulate creativity.

In sum, this study contributes novel insights into the neural correlates of creative idea generation. It also demonstrates that using an immersive VR environment as a support tool for creative idea generation tasks could significantly enhance the user's experience of immersion, flow, and enjoyment, which are all factors that have been associated with creative thinking in prior research. In this study, however, we did not examine the relationship between these enhanced factors, brain activity, and subjects' creative performance. In future research, we aim to employ subjective and objective metrics (such as expert ratings or analysis of semantic distance [19]) to evaluate the quality of the generated creative ideas and establish the relationships between brain activity patterns and creative thinking performance. More research is required to further address contradicting findings in past research, particularly the role of alpha band activity, in different creativity tasks supported by digital tools [13, 30, 31].

5 CONCLUSION

In this study, we investigated the effectiveness of immersive VR environments for supporting creative idea generation using subjective self-reports and objective measures of brain activity. The results showed that the participants felt more immersed and experienced greater time distortion and enjoyment when they worked on a problem-based idea generation task in a 3D VR environment compared to a 2D computer screen. Additionally, the analysis of the EEG brain signals revealed a significant increase in frontal beta and gamma oscillations during the 3D VR condition, which have been associated with enhanced sensory processing and information integration in the previous neuroscience literature. To conclude, VR systems hold potential for enhancing creative idea generation as they offer immersive environments that promote the experience of flow. Using real-time measures of brain activity, it is possible to develop VR creativity support tools that monitor a user's mental state and provide timely cues and nudges that stimulate creativity.

6 APPENDIX

Previews of the videos that have been used for the experiment can be found at the following links:

Scenario 1: <https://www.storyblocks.com/video/stock/360vr-video-of-an-incline-in-a-forest-on-a-sunny-day-bfxhssjgrjxogo4hj>

Scenario 2: <https://www.storyblocks.com/video/stock/360-vr-video-waterscape-and-landscape-of-mauritius-ocean-shallow-water-and-black-rocks-on-the-coast-clouds-sailing-over-the-island-rujhglr3xj0qdlz57>

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